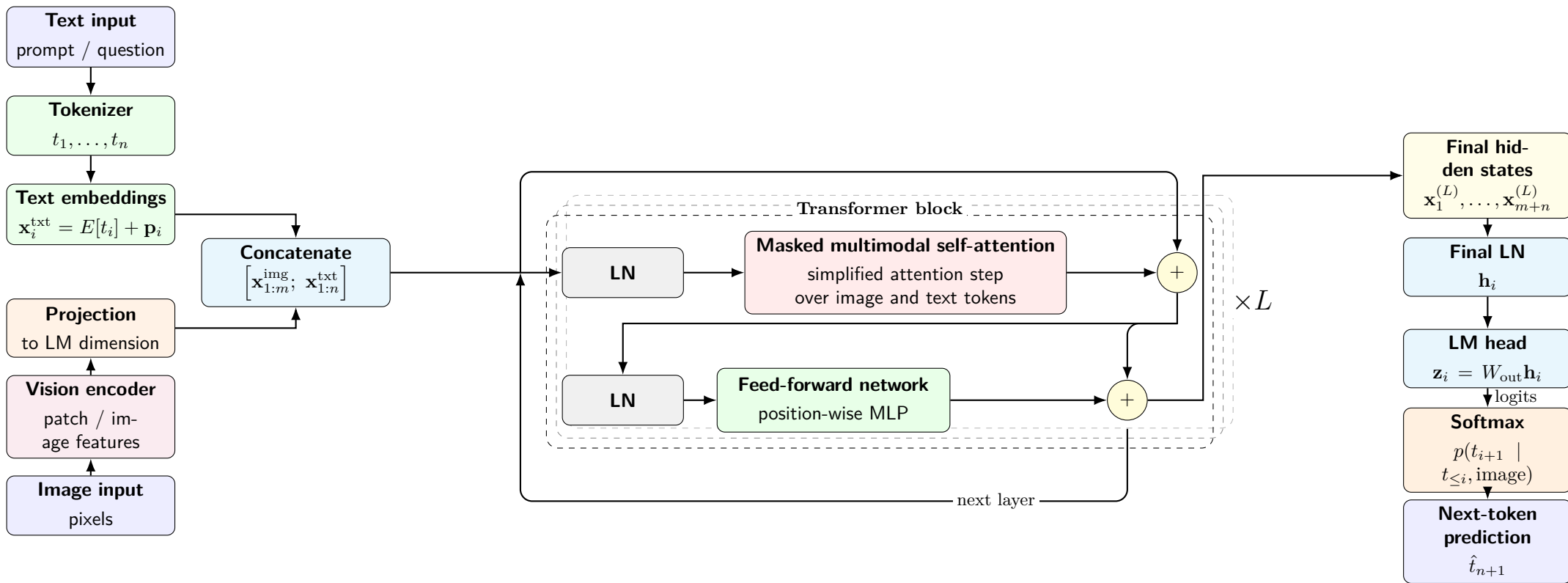


Vision–Language Transformer: Complete Forward Pass



Modality-specific front ends

Text is tokenized and embedded. Images are converted into visual features, then projected into the same hidden dimension as text-token embeddings.

One pre-norm transformer block

$$\tilde{\mathbf{x}}^{(\ell)} = \mathbf{x}^{(\ell)} + \text{Attn}(\text{LN}(\mathbf{x}^{(\ell)})), \quad \mathbf{x}^{(\ell+1)} = \tilde{\mathbf{x}}^{(\ell)} + \text{MLP}(\text{LN}(\tilde{\mathbf{x}}^{(\ell)}))$$